KARTIKEYA SHARMA

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SUMMARY

Security and applied-ML engineer (5+ years) specializing in threat detection, anomaly detection, and adversarial AI. Owns detection systems end-to-end from dataset construction through model deployment, with published research on graph-based intrusion detection and production tooling built on LLM and embedding systems.

EXPERIENCE

Senior Associate Information Security Engineer, ML & Analytics Jul 2023 – Present

Equinix, Inc. • Remote, US

- Designed and built a graph neural network detection pipeline that retrospectively identifies DDoS activity from raw edge firewall logs, spanning three architectures (flow-based GNN, heterogeneous U-Nets, and an unsupervised encoder-decoder).
- Developed an unsupervised anomaly detection system over endpoint application-install data from CrowdStrike and Tenable, applying K-Means clustering and cosine similarity to surface anomalous software across [N] endpoints that EDR tooling misses.
- Delivered a semantic string-matching service backed by a background processing queue for large files, collapsing a recurring multi-day reconciliation task to under an hour (95%+ faster) while retaining human verification of results.
- Built an IOC enrichment tool that aggregates multiple threat-intelligence sources into a single bulk view, scaling analysts from one-at-a-time lookups to thousands of indicators per run.

Data Scientist Jul 2019 – Sep 2021

Veada Industries (Lippert Components) • New Paris, IN

- Built and deployed (AWS Elastic Beanstalk) a Flask/jQuery/REST project-management web app replacing legacy software and manual processes, cutting manufacturing lead time by 50%.
- Trained a Scikit-learn/NLTK classifier to auto-categorize incoming orders, informing labor- and material-planning decisions.

RESEARCH, TALKS & PUBLICATIONS

- Adversarial Techniques for Bypassing GNN-Based Network Defense | SaintCon 2025
- Graph Neural Network–Based DDoS Protection for Data Center Infrastructure | PNSQC 2025
- GPT as Ally: Improving Cybersecurity Workflows with GPT & Embedding APIs | BSides Austin 2024
- Graphing the Insider: Innovative Applications of GNNs in Insider Threat Detection | BSides Portland 2024

EDUCATION

- **M.S. Computer Science**, University of Oregon (2021–2023)

GPA: 4.08

Coursework: Computer & Network Security, Machine Learning, Intro to AI, Multi-Agent Systems, Data Science, Distributed Systems, Algorithms.

- **B.A. Computer Science & Accounting**, Goshen College (2015–2019)
  GPA 3.64

SERVICE

- Judge, Google I/O Hackathon 2026 (Cerebral Valley with Google DeepMind)
- CFP reviewer for 5 security and engineering conferences

SKILLS

- **Languages:** Python, SQL, JavaScript, C/C++, Bash
- **ML & AI:** PyTorch, DGL, Scikit-learn, NLTK, pandas, NumPy; GNNs, anomaly detection, adversarial ML, LLM & AI-agent systems, dataset construction
- **Security & Cloud:** Google Cloud, anomaly/threat detection, CrowdStrike, Tenable, IOC/threat-intel enrichment, Elasticsearch, SOAR, AWS
- **Data:** SQL Server, PostgreSQL, MySQL, Apache ECharts, Power BI